

# Vitchakorn Poonyakanok, MD, MS

*Physician & Clinical Data Analyst*

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## Professional Profile

Physician-technologist working at the intersection of frontline clinical practice and data science — a bridge between the two worlds. I turn clinical and public-health problems into working software and analytics: a deployed altitude-risk decision-support app ([altit.app](#)), field outbreak investigations, predictive clinical models, and edge-AI health devices. I bring clinical credibility to data and engineering teams, and engineering rigour (ML, data governance, interoperability, cybersecurity) to health systems.

## Education

- Aug 2026 – May 2028 **M.S. in Health Care Analytics & Information Technology (MSHCA)**, *Heinz College, Carnegie Mellon University*, Pittsburgh, PA, USA, Royal Thai Government (Department of Disease Control, DDC) scholarship  
Expected graduation: May 2028. Focus: health system analytics, data governance, and information systems.
- Aug 2025 – Jun 2026 **M.S. in Digital & AI Technologies in Health Systems (DigiHealth)**, *Faculty of Medicine & Faculty of Engineering, Chulalongkorn University*, Bangkok, Thailand  
Focus: digital health architecture, clinical AI applications, interoperability, and medical databases.  
Independent study: *Development and Evaluation of an AI-Assisted Altitude Extraction System from Travel Itineraries for Travel Medicine* (advisor: Assoc. Prof. Watcharapong Piyaphanee, M.D.).
- Jun 2024 – Jul 2026 **Residency Training in Preventive Medicine (Travel Medicine)**, *Institute of Preventive Medicine, DDC, Ministry of Public Health (MOPH)*, Nonthaburi, Thailand  
Focus: preventive medicine, travel risk assessment, vaccinology, and health security.  
On planned academic leave from Aug 2026 to undertake the MSHCA at Carnegie Mellon; residency to resume on return.
- Apr 2013 – Jul 2019 **Doctor of Medicine (MD)**, *Faculty of Medicine Ramathibodi Hospital, Mahidol University*, Bangkok, Thailand  
Clinical rotations across Internal Medicine, Surgery, Pediatrics, OB-GYN, and Emergency Medicine.  
International clinical elective via the IFMSA SCOPE exchange — Hospital Clínico Universidad de Chile, Santiago (2017).
- 2010 – 2013 **Upper Secondary School (Science-Mathematics Program)**, *Triam Udom Suksa School*, Bangkok, Thailand

## Work Experience

- Nov 2024 – Jul 2026 **Medical Physician, Professional Level — Preventive & Travel Medicine**, 📄 *Office of Disease Prevention and Control 6 (ODPC 6), DDC, MOPH*, Pattaya, Chon Buri, Thailand
- Deliver pre-travel risk assessment, yellow-fever vaccination/certification, and prophylaxis counselling; apply self-built clinical tools (e.g. altitude-risk analysis) at the point of care.
  - Serve as site physician for the WHO Enhanced Gonococcal Antimicrobial Surveillance Programme (EGASP), contributing standardized surveillance data.
- On academic leave from Aug 2026 for the MSHCA at Carnegie Mellon.

- Jun 2022 – **General Practitioner**,  *Bamrasnaradura Infectious Diseases Institute*, Nonthaburi, Thailand
- May 2024
- Led outpatient and Acute Respiratory Infection (ARI) clinic services through major COVID-19 and mpox outbreak responses (peak ~50 outpatients/day per physician).
  - Analyzed 32,084 ARI-clinic visits (26,669 patients; Mar 2022 – Aug 2023) in Python — arrival hour was the dominant bottleneck ( $r = -0.41$ ; ~121 min mean service time for 07:00 arrivals vs ~44 min at 14:00; weekends/holidays 15–21 min slower, ANOVA  $p < 0.001$ ) — and presented scheduling recommendations to department leadership.
  - Delivered vaccinology counselling and designed clinical safety protocols for staff.
- Jul 2020 – Apr 2022 **General Practitioner (Intern)**,  *Si Songkhram Hospital*, Nakhon Phanom, Thailand
- Managed general, emergency, and inpatient medical services at a 60-bed community hospital.
- Sep 2019 – Jul 2020 **General Practitioner (Intern)**,  *Nong Khai Hospital*, Nong Khai, Thailand
- Completed intern clinical rotations in emergency medicine, pediatrics, surgery, and OB-GYN at a ~350-bed provincial general hospital.

## Publications & Outbreak Investigations

- Peer-reviewed **An Investigation of MMR-Related Mumps Cluster Following Immunization Among Practical Nursing Students, Bangkok, Thailand, 2024**. S. Lertsakulbunlue, D. Srithammavong, K. Tepsittha, H. Kanjanasombut, V. Poonyakanok, et al. *Journal of Tropical Medicine* 2025;2025:9974081. (5th of 16 authors; field outbreak investigation)
- In preparation **An Investigation of Mumps Outbreak in a Secondary School, Saraburi Province, Thailand, August–September 2025**. T. Prasongsil, W. Saowong, V. Poonyakanok, et al. Manuscript in preparation. (co-author; DDC Joint Investigation Team field investigation — retrospective cohort & Firth’s logistic regression)
- Surveillance evaluation **Evaluation of the Syphilis Surveillance System**, ODPC 6, Pattaya, Chon Buri, Thailand (Jan–Jun 2025). Assessed the national Digital Disease Surveillance (DDS) system’s attributes — sensitivity, predictive value positive, timeliness, and representativeness — and recommended improvements. (co-investigator; preventive-medicine residency study)
- Surveillance report **Preliminary Investigation and Rapid Response to a Methanol Intoxication Outbreak Linked to Illegal Herbal Liquor in Bangkok, August 2024**. C. Jitpeera, W. Rongdech, K. Raruenroeng, V. Poonyakanok, et al. *Weekly Epidemiological Surveillance Report (WESR)*, 2024. (co-author; rapid outbreak response)

## Honors & Awards

- Jul 2026 **2nd Place (Silver Award), Track B — Digital Health Innovation Competitions, BRIDGE-AI Summit 2026**, Faculty of Medicine, Chulalongkorn University (with Weill Cornell Medicine) — 2nd of 15 teams in the Medical AI category (60 graduate & medical-resident teams across four categories), for the AI-assisted altitude-extraction work behind [altit.app](#).

## Applied AI & Technical Training

- Apr – Jun 2026 **Super AI Engineer Season 6, Artificial Intelligence Association of Thailand (AIAT)**, Thailand, Advanced AI Level 2 — selected as one of 157 from 10,457 applicants
- **Coffee Chain Demand Forecasting Hackathon**: Private Kaggle Rank #1 / 2nd Runner-up overall (of 30 teams) — leakage-aware time-series pipeline with gradient boosting and robust cross-validation.
  - **Parasite Eggs Object Detection Challenge**: Ranked 16/166 — cascaded microscopy detector using SAHI, YOLOv8, and a DINOv2 classification head.
  - **Edge-AI Intelligent Transport System Hackathon**: Co-Honorable Mention — real-time CCTV edge analytics on Jetson Nano (YOLO/ONNX Runtime).
  - **WellSense AIoT Product Hackathon**: Academic Popular Vote — wearable running-posture monitor (Arduino Nano 33 BLE Sense, TinyML/Edge Impulse, BLE, React dashboard).
  - **Agentic Pipeline Hackathon (2026 Thai Election OCR)**: Ranked 102/571 — OCR pipeline for hand-written election reports.

## Selected Projects

### Deployed tools

- **DDS Chatbot** ([live demo](#)) — guardrailed Thai disease-surveillance LLM assistant for the DDC: answers epidemiology questions via LLM tool-calling over an aggregate, PII-free data mart, behind a six-layer, PDPA-aligned guardrail stack. Hardened via a 753-turn authorized red-team and stress-testing session — prompt-injection, system-prompt-extraction, and destructive-query attacks — with live hot-fixes cutting attack success from ~25% to ~1% in-session; 99.8% pass on a 3,819-run secure regression benchmark; presented to the DDC Division of Epidemiology (Jun 2026).
- **Altitude Itinerary Analyzer** ([altit.app](#)) — full-stack clinical decision support web app that parses and analyzes multi-modal travel itineraries for high-altitude sickness risk assessment; 84.9% altitude-extraction accuracy within  $\pm 100$  m on text itineraries in the thesis benchmark (2nd Place, BRIDGE-AI Summit 2026).
- **Thai HospSearch** ([hoscode.vercel.app](#)) — in-browser lookup over ~174K MOPH-coded records (44K health facilities, 42K schools, 77K villages) across Thailand's 13 health regions, 77 provinces, and 7,436 subdistricts, for administrative coding and records checking; auto-refreshed weekly from MOPH APIs.
- **Hajj Menstrual Delay Planner** ([hajj-period-planner.vercel.app](#)) — clinical utility for planning pharmaceutical menstrual suppression for female Hajj pilgrims.

### Data science

- **Cirrhosis Survival Prediction** ([Kaggle](#)) — multi-class survival model using ensemble learning (XG-Boost/LightGBM).
- **Outpatient Waiting Time Analysis** ([Kaggle](#)) — ANOVA and Random Forest on hospital logs to pinpoint bottlenecks and forecast wait times.
- **Sichuan-Yunnan Altitude & Transit Map** ([GitHub](#)) — QGIS geospatial visualization for Acute Mountain Sickness risk along the Sichuan-Yunnan highway.

### Dashboards

- **Tableau: PM2.5 & Mental Health Dynamics** (with Interrupted Time Series), **Yellow Fever & Malaria Prevention**, and **Thai EPI Vaccine Dashboard**.

## Certifications

### Security & Cloud

- TPQI × Huawei — **AI Literacy Power Users & Cloud Developer Level 5** — 2026.
- **ISC2 Certified in Cybersecurity (CC)** — Apr 2026–Mar 2029.
- Huawei Cloud HCCDA — **Tech Essentials & AI** — Apr 2026–Apr 2029.
- **Cybersecurity Foundation Course**, Thailand National Cyber Academy — 2024.
- **Google Cybersecurity Professional Certificate** — 2024.

### Data & AI

- **Super AI Engineer** — **AI Practitioner Certification (Level 1)**, AIAT — 2026; includes **Foundation AI (Theory)**, **From Data to Insight**, and the **FahMai RAG & Agentic Pipeline** hackathons.
- **Google AI Essentials V1** — 2024.
- **Certified Professional Data Analyst**, DataRockie Bootcamp 9 — 2024.
- **Google Advanced Data Analytics Professional Certificate** — 2023.
- **Data Visualization with Tableau Specialization**, UC Davis — 2023.
- **Excel Level Up Update 2021**, SkillLane — 2023.
- **Google Data Analytics Professional Certificate** — 2023.

### Medical & Research

- **CITI Program Biomedical Researchers** — Jan 2025–Jan 2029.
- **Introductory Course in Epidemiology & Biostatistics**, DDC MOPH — 2024.
- **ISTM Review Course**, International Society of Travel Medicine — 2024.
- **Diploma in Clinical Epidemiology**, Chiang Mai University — Sep–Dec 2022.
- **Diploma in Clinical Statistics**, Chiang Mai University — Feb–May 2022.

## Licensure & Registration

**Medical License**, Thai Medical Council — full registration since 2019.

**Tour Leader License**, Department of Tourism, Thailand — 2025.

**Authorized Seafarers' Medical Examiner**, Office of Maritime Labour, Thailand — 2026; authorized to issue seafarers' medical certificates.

## Professional Memberships

**HIMSS** (Healthcare Information and Management Systems Society) — member since 2026.

## Technical Skills

**Programming & Statistics:** Python, R, SQL, STATA

**Machine Learning & Data Science:** Scikit-Learn, gradient boosting & stacked ensembles (XGBoost, LightGBM, CatBoost), AutoGluon (AutoML), Chronos (time-series), PyTorch, YOLO, OpenCV, Pandas, NumPy, Jamovi

**LLM & Generative AI:** Claude, GPT, Gemini, GLM, Grok, and ThaiLLM APIs; RAG and agentic pipelines

**Visualization & Geospatial:** Tableau, Looker, Power BI, Matplotlib, Seaborn, QGIS, Adobe Photoshop, Inkscape

**Software, Cloud & Hardware:** Git, Firebase, Google Cloud Platform (GCP), Huawei Cloud, Arduino, TinyML, UML

**Environments & HPC:** Colab, Jupyter, Anaconda, Linux, LANTA HPC

**Clinical Data & Interoperability:** data interoperability, data governance, medical record systems

## Languages & Interests

**Languages** Thai (Native); English — Proficient (TOEFL iBT 91/120, Duolingo English Test 120/160).

**Medical** Travel medicine, preventive vaccinology, emerging infectious diseases.

**Technical** Public health informatics, clinical machine learning, geospatial disease mapping, healthcare cybersecurity.

**Personal** Scuba diving (PADI Rescue Diver), photography, trekking, and travel.